

Amir Mollaei

Amir Hossein Mollaei Khass

Ph.D. Student in Mechanical Engineering

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Research Profile

I study safe active perception: how robots acquire useful observations while navigating partially observed environments. My research connects risk-aware planning and control with online 3D Gaussian Splatting (3DGS), efficient next-best-view selection, and distributed coordination with private maps. I develop and evaluate these methods in simulation and on physical mobile robots.

Research interests: Safe active perception; robot learning; next-best-view planning; active scene learning; uncertainty-aware navigation; differentiable 3D mapping; information-theoretic sensing; distributed multi-robot optimization.

Education

Lehigh University

2023–Present

Ph.D. student, Mechanical Engineering

Bethlehem, PA, USA

Autonomous and Intelligent Robotics Lab (AIRLab).

Advisor: Prof. Nader Motee (motee@lehigh.edu).

Passed the doctoral general examination in March 2026.

Lehigh University

2025

M.S., Mechanical Engineering

Bethlehem, PA, USA

Sharif University of Technology

2023

B.Sc.

Tehran, Iran

Research Appointment

Graduate Research Assistant

Lehigh University

Autonomous and Intelligent Robotics Lab (AIRLab); advisor: Prof. Nader Motee

Develop methods for risk-aware control, active sensing, and distributed view selection over 3DGS maps. Build mapping and control pipelines in Isaac Sim and Habitat-Sim, with physical Ackermann-robot validation; detailed projects follow below.

Teaching Experience

Teaching Assistant, Lehigh University

Courses: *Convex Optimization*; *Control Systems*.

Technical Skills

Programming: Python, C++, MATLAB.

Robotics and simulation: ROS/ROS2, Isaac Sim, Isaac Lab, Habitat-Sim, MuJoCo.

Methods: 3DGS, control barrier functions, convex optimization, quadratic programming, Fisher information, expected information gain, Riemannian optimization, consensus ADMM.

Software and tools: PyTorch, Git, Linux.

Robot and sensing platforms: Kinova Gen3 7-DoF manipulator, Ackermann-steered mobile robots, RGB-D cameras.

Publications and Manuscripts

Author names follow the corresponding publication records; my name is bolded.

Google Scholar snapshot supplied October 2026: 29 citations; h-index 2; i10-index 1.

Conference Papers — Published or Accepted

- [1] **Conflict-Aware Active Perception and Control in 3D Gaussian Splatting Fields via Control Barrier Functions**
Amirhossein Mollaei Khass, Athanasios Cosse, Vivek Pandey, and Nader Motee.
65th IEEE Conference on Decision and Control (CDC), 2026. **Accepted; invited session paper.** [arXiv:2605.20566](#)
- [2] **Efficient Greedy Algorithms for Feature Selection in Robot Visual Localization**
Vivek Pandey, **Amirhossein Mollaei**, and Nader Motee.
European Control Conference (ECC), 2026. [arXiv:2511.20894](#)
- [3] **Active Next-Best-View Optimization for Risk-Averse Path Planning**
Amirhossein Mollaei Khass, Guangyi Liu, Vivek Pandey, Wen Jiang, Boshu Lei, Kostas Daniilidis, and Nader Motee.
IEEE International Conference on Robotics and Automation (ICRA), 2026. [arXiv:2510.06481](#)

Workshop Contribution

- [4] **SemSafe-3DGS: Semantic Risk-Aware Active Navigation in Uncertain 3D Gaussian Splatting Maps**
Amirhossein Mollaei Khass, Athanasios Cosse, and Nader Motee.
SeMaNa Workshop, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. **Accepted for presentation; presented September 2026.** [arXiv:2609.19330](#)

Manuscripts Under Review

- [5] **TRACE: Privacy-Preserving Next-Best-View Selection over Distributed 3D Gaussian-Splat Maps**
Amirhossein Mollaei Khass, Athanasios Cosse, Qiyu Sun, and Nader Motee.
Preprint, 2026. **Under review at ICRA 2027.** [arXiv:2610.00822](#)
- [6] **AGILE-GS: Anchor-Guided Fast Next-Best-View Selection for Active 3D Gaussian Splatting**
Amirhossein Mollaei Khass and Nader Motee.
Preprint, 2026. **Under review at IEEE/CVF WACV 2027.** [arXiv:2609.34176](#)
- [7] **LiTe-GS: Oracle-Efficient Next Best View Selection for 3D Gaussian Splatting**
Vivek Pandey, **Amirhossein Mollaei Khass**, and Nader Motee.
Preprint, 2026. **Under review at IEEE/CVF WACV 2027.** [arXiv:2609.30393](#)
- [8] **Splat-CBF: Safe Next-Best-View Control in 3D Gaussian-Splat Maps**
Amirhossein Mollaei Khass, Athanasios Cosse, and Nader Motee.
Preprint, 2026. **Under review at ICRA 2027.** [arXiv:2609.23100](#)

Additional Preprint

- [9] **Multi-Agent Next-Best-View Optimization for Risk-Averse Planning**
Amirhossein Mollaei Khass, Vivek Pandey, Guangyi Liu, Athanasios Cosse, Emrah Bayrak, and Nader Motee.
Preprint, 2026. [arXiv:2606.04158](#)

Research Projects and Contributions

AIRLab, Lehigh University. Project descriptions summarize methods and evaluations from the corresponding coauthored papers.

Splat-CBF: Safe Motion and Active Scene Learning

- Developed a unified controller directing the camera toward informative views while enforcing collision avoidance through a hard control barrier constraint.
- Formulated a fast, smooth composite safety barrier from Average Value-at-Risk (AV@R) clearances, replacing obstacle-wise constraints with one aggregate constraint.
- Coupled trajectory-relevant Fisher information with a soft perception barrier in a quadratic program. Adapted the slack penalty using perception-relaxation history and proximity to uncertain regions.
- Integrated online 3DGS mapping and operational-space safety filtering for a Kinova Gen3 7-DoF manipulator in Isaac Sim; evaluated view-selection alternatives with collision-free experimental trials.
- Validated safe active perception on a physical Ackermann-steered mobile robot, maintaining a positive safety barrier throughout the reported traversal.

Conflict-Aware Active Perception and Control

- Derived a risk-aware safety control barrier function directly from uncertainty in the 3D Gaussian map, using an AV@R collision metric rather than a thresholded occupancy representation.
- Combined next-best-view information objectives and camera-orientation perception barriers with hard safety constraints in a common quadratic program.
- Resolved sensing–safety conflicts using slack-relaxed perception constraints, reducing perception demands when required by safety.

Risk-Averse Planning with Active Next-Best-View Optimization

- Coupled risk-averse path refinement and active sensing through an online 3DGS map, focusing observations on the robot's upcoming trajectory.
- Refined reference paths using AV@R risk statistics and local A* search over risk-filtered grid points to generate conservative, short-horizon trajectories.
- Optimized camera poses through a Riemannian gradient method with a proximity-weighted Fisher-information objective and a mask around the planned trajectory.

SemSafe-3DGS: Semantic Risk-Aware Navigation

- Incorporated semantic attributes into Gaussian-map risk assessment using class-dependent weights, allowing geometrically similar obstacles to induce different safety responses.
- Aggregated weighted AV@R clearances into a composite safety barrier and paired it with trajectory-relevant active perception in a unified CBF-QP.
- Evaluated semantic-dependent trajectory adaptation and physical robot execution under Ackermann dynamics; prepared a [workshop poster](#) for the IROS presentation.

AGILE-GS: Fast Next-Best-View Selection

- Proposed anchor-guided view selection that separates information search from choosing an executable candidate camera pose.
- Optimized a virtual anchor on $SE(3)$ using Riemannian gradient ascent; used viewing-geometry signatures and greedy ridge-leverage selection to form a nonredundant shortlist without rendering candidate views.
- Developed AGILE-GS and AGILE-GS+ to avoid candidate Fisher evaluations or restrict them to the shortlist. Reported one to two orders of magnitude lower selection latency in benchmark and closed-loop acquisition experiments.

Research Projects and Contributions (continued)

TRACE: Distributed View Selection with Private Maps

- Developed a protocol for joint next-best-view decisions while each robot retains its observations and Gaussian map parameters locally.
- Reconstructed expected information gain and its orientation gradient using depth-binned front-transmittance and behind-radiance aggregates with pose derivatives. Message size does not grow with map size.
- Characterized exact reconstruction and bounded error when behind-splat depth bins mix hits from different robots. Across 100 Habitat-Sim decisions, selected views achieved **97.9% of centralized information gain**, with headings within 15° of the centralized choice in **83.3%** of cases.

Multi-Agent Risk-Averse Next-Best-View Optimization

- Formulated distributed information acquisition across robots maintaining separate local 3DGS maps, with sensing objectives restricted to risk-relevant zones along planned trajectories.
- Applied consensus ADMM over a communication graph, exchanging candidate viewpoints, trajectory descriptors, and scalar information-gain contributions rather than raw maps or sensor observations.
- Evaluated multiple team sizes in Gibson environments, comparing mapping quality, trajectory safety, and communication requirements against centralized coordination.

LiTe-GS: Oracle-Efficient Active View Selection

- Coauthored a randomized view-selection method that reduces costly information-gain evaluations by scoring subsets of candidates instead of the full pool.
- Established expected $O(M \log(1/\epsilon))$ oracle complexity, independent of selection cardinality, with a controllable trade-off between computational cost and approximation quality.
- Evaluated reconstruction quality and Fisher-oracle cost on Blender and Mip-NeRF 360 against information-based view-selection baselines.

Efficient Feature Selection for Visual Localization

ECC 2026; publication [2]

Coauthored two fast, memory-efficient greedy algorithms for selecting visual features informative for localization over a prediction horizon. The work addresses redundant or uninformative image features and the trade-off between computational efficiency and localization accuracy.

Research Software and Demonstrations

- **Isaac Sim interface:** Online 3DGS mapping, view selection, and safety filtering; manual, keyboard teleoperation, and automatic acquisition modes.
- **Public research code:** Risk-Averse Next-Best-View Selection; Conflict-Aware Active Perception, with an accompanying project website.
- **Research demonstrations:** Online portfolio with methods and videos for risk-averse planning, Splat-CBF, TRACE, AGILE-GS, LiTe-GS, and multi-agent NBV.

Conference and Poster Presentations

Northeast Robotics Colloquium (NERC), Princeton University

October 3, 2026

Perception Control Barrier Function Approach to Next-Best View-Action Planning in 3D Gaussian Maps.

Coauthors: Athanasios Cosse and Nader Motee. **Accepted for poster presentation.**

SeMaNa Workshop, IEEE/RSJ IROS 2026

September 2026

Presented *SemSafe-3DGS: Semantic Risk-Aware Active Navigation in Uncertain 3D Gaussian Splatting Maps.*

European Control Conference (ECC), Reykjavik, Iceland

July 2026

Presented research on efficient greedy feature selection for robot visual localization.

IEEE ICRA, Vienna, Austria

June 2026

Presented *Active Next-Best-View Optimization for Risk-Averse Path Planning.*